

STATIC GK — Computer

Batch 12 of 14 · Source: `static\_gk\_combined\_raw\_extraction.md` #12 Science & Technology/Computer (Lucent GK 2026 Ch.12 "Computer", pp.488-498) · PYQ bucket "11 Computer" — the smallest owned bucket, only 4 Qs, and none of them test classic computer-basics content (they're 2025 current-affairs: Quantum Year, an AI model, modulation theory, AI in healthcare) · ★ = tested in a real past SSC paper · this is inherently a reference chapter — most entries below are book-backed but not yet PYQ-tested in this book's sampled papers

1 History & Generations

Item	Fact
Pascaline	First mechanical calculator, Blaise Pascal, 1642
Charles Babbage	Conceived an automatic calculator, 1833 — "Father of the modern computer"
Ada Lovelace	First computer program, Babbage's student
Mark-I	First mechanical computer, Howard Aiken, 1937
ENIAC	World's first electronic computer, Eckert & Mauchly, 1946 — first electronic digital computer of the world
EDVAC	Von Neumann, 1951 — introduced Stored Program concept and the Binary System

Generation	Period	Component	Examples
I	1940-52	Vacuum Tube	EDSAC, EDVAC, UNIVAC
II	1952-64	Transistor	IBM-700/1401/1620
III	1964-71	Integrated Circuit	IBM-360/370
IV	1971-	Large-Scale IC	Apple, DCM
V	—	Optical Fibre, AI	—

FORTRAN = first programming language; PROLOG = 5th-generation language. Computer types by size: Micro (PC), Mini, Mainframe (100+ simultaneous users), Super (fastest/costliest), Quantum (uses Q-bits).

2 India-Specific & Supercomputer Facts

Item	Fact
Siddharth	First computer made in India, by ECIL
Silicon Valley of India	Bengaluru
CRAY K1S	World's first supercomputer, Cray Research Co., USA
Deep Blue	IBM supercomputer that defeated Garry Kasparov (chess)
Seymour Cray	"Father of the Supercomputer"
PARAM-10000	C-DAC Pune, 28 March 1998, 1 billion calculations/sec
PARAM KAMRUPA	Inaugurated by President Droupadi Murmu, IIT Guwahati, 13 Oct 2022
FLO SOLVER	First Indian supercomputer, National Aerospace Laboratories, Bangalore
Simputer	Palm-sized touch-screen computer, IISc Bangalore
Linus Torvalds	Creator of Linux; first version completed 17 Sept 1991, released under GNU General Public Licence

3 Hardware, Memory & Units

Item	Fact
Bit/Byte	8 bits = 1 byte (256 symbols); 4 bits = 1 nibble
Data units	1024 B=1 KB, 1024 KB=1 MB, 1024 MB=1 GB, 1024 GB=1 TB, 1024 TB=1 PB
CPU parts	ALU (arithmetic/logic), CU (control), Registers (fastest memory, made of flip-flops)
Clock speed	Measured in MHz/GHz; 1 GHz = 1000 MHz
RAM vs ROM	RAM = volatile, read/write; ROM = non-volatile, fixed at manufacture; PROM = programmable once
Logic gates	Three basic: AND, OR, NOT

OMR / MICR	OMR reads pencil marks (competitive exams); MICR reads magnetic-ink cheque numbers (banks)
Printers	Dot-matrix, ink-jet, laser (laser = fastest)

4 Key Abbreviations & Miscellaneous Facts

December 2 = Computer Literacy Day · India's New Computer Policy, 1984 · First computer installed in India = Bangalore GPO, 16 Aug 1986 · Three computer types: Digital, Analog, Hybrid · Most-used OS: Windows (PC), Android (smartphones — highest overall), iOS (tablets); most supercomputers run Linux · Bug = a computer error · Y2K = "Millennium Bug" · J.S. Kilby developed IC chips · C-DAC established Pune, 1988 · Vellanad (Thiruvananthapuram, Kerala) = India's first fully computerised village.

5 Keyboard Shortcuts (reference table)

Key	Function	Key	Function
Ctrl+A	Select All	Ctrl+N	New document
Ctrl+B	Bold	Ctrl+O	Open document
Ctrl+C	Copy	Ctrl+S	Save
Ctrl+X	Cut	Ctrl+P	Print
Ctrl+Z	Undo	Ctrl+Y	Redo
Ctrl+I	Italics	Ctrl+U	Underline
Ctrl+E/L/R/J	Centre/Left/Right/Justify alignment	Ctrl+K	Hyperlink
F1	Help	F7	Spell Check
F2	Rename	F12	Save As
Alt+Tab	Switch program	Alt+F4	Close window

6 PYQ-Tested Facts (bucket "11 Computer" — all current-affairs, outside the classic chapter's scope)

Fact	Detail
International Year of Quantum Science and Technology ★	Declared by the UN for 2025 ★ — marks the 100th anniversary of quantum mechanics — NOT WTO/OECD/NATO (tested distractors)
Gemini AI model ★	Developed by Google DeepMind ★ — NOT Apple/Amazon/IBM (tested distractors; Amazon's own chatbot is "Rufus")
AM/FM/PM modulation ★	In Amplitude Modulation, bandwidth of the modulated signal = twice the modulating-signal frequency ★; Phase Modulation becomes equivalent to Frequency Modulation when the modulating signal is integrated before the phase modulator ★ — both tested statements correct
AI in healthcare ★	Improves healthcare mainly by providing accurate diagnosis through data analysis ★ (early symptom detection, reduced human error in imaging, personalised treatment) — NOT replacing doctors/reducing infrastructure/eliminating records (tested false options)

DOUBT The raw extraction itself flags several unverified attributions in this chapter: "Soul of New Machine" credited to Ted Nelson winning a Pulitzer (usually attributed to Tracy Kidder's *The Soul of a New Machine*); PARAM-10000's credited scientist printed as "Dr. Vijay P. Bhaskar" (usually cited as Dr. Vijay P. Bhatkar). Neither is touched by this batch's PYQ set.

A.1 — PYQ citation index

Fact tested	Exam · Date · Shift
UN declared 2025 International Year of Quantum Science and Technology	SSC CHSL TIER-1 17/11/2025 (Shift-03)

Gemini AI model developed by Google	SSC PHASE XIII 01/08/2025 (Shift-02)
AM/FM/PM modulation statements	SSC CPO TIER-1, 09/12/2025 (Shift-01)
AI improves healthcare via accurate diagnosis	SSC PHASE XIII 24/07/2025 (Shift-01)

Coverage: all 4 questions in bucket 11 are represented above; none map to the classic Computer chapter's generations/hardware/shortcuts content, which remains untested in this sample.

A.2 — PYQ-book-only facts

All of §6 is PYQ-book-only relative to the raw extraction — the Lucent "Computer" chapter is a 2026-edition textbook chapter on computer fundamentals (history, hardware, shortcuts, glossary) and does not cover 2025 current-events like the UN's Quantum Year declaration, the Gemini AI model, or AI-in-healthcare applications. Sourced to the PYQ book's own explanations.

A.3 — Doubts carried forward from the raw extraction

See the DOUBT box above (Ted Nelson/Pulitzer attribution; PARAM-10000 credited-scientist name) — both are raw-extraction-flagged printing/attribution uncertainties, not touched by this batch's PYQ set.